

Predicting Social Media Content Virality Across Instagram, YouTube, and TikTok

A Multi-Platform, Fairness-Aware Classification Study

CSCI-4930 Machine Learning — Spring 2026 — University of Colorado Denver

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GitHub: github.com/taru01ok/virality-prediction-ml-2026

Data: drive.google.com/drive/folders/1aWALtnJTxLUQHyaCJNKfxW6wwZprxgo

1. Introduction

Social media virality determines whose content reaches mass audiences, affecting health information, purchasing behavior, and public discourse. This project builds a machine learning classifier to predict, before a post is published, whether content on Instagram, YouTube, or TikTok will go viral. Virality is defined as a post exceeding the top 20th percentile engagement rate for its platform. The model uses only pre-publication features: caption sentiment, hashtag count, posting time, content category, caption length, and platform indicator.

Two gaps motivate this work. First, most existing research predicts virality using post-publication signals unavailable before posting. Second, no published work has audited fairness across platforms or content communities. This project addresses both gaps by building six classifiers and conducting a thorough fairness audit.

2. Related Work

Adhikari et al. (2025, IEEE INCET) achieved 95% accuracy with a stacking ensemble but relied on post-publication signals without fairness auditing. Silva and Rajapaksha (2023, ICATC) predicted Instagram virality with Random Forest at 85% accuracy — Instagram-only. Vallet et al. (2015, ACM CIKM) used Gradient Boosted Trees for YouTube and Twitter but required post-publication data. Roring et al. (2026, IRJMS) validated timing and trending audio as virality drivers but lacked standard ML metrics. Noor and Rauf (2025) found ensemble methods consistently outperformed baselines for news virality. No prior work combines pre-publication features across all three platforms or evaluates fairness.

3. Data

3.1 Datasets

Three Kaggle datasets were used. The Instagram Analytics Dataset (29,999 posts, 23 columns) includes engagement metrics, content category, caption length, hashtag count, and posting time

pre-computed. The YouTube Trending Video Statistics US dataset (40,949 videos) includes title, tags, views, likes, comment count, publish time, and category ID. The TikTok dataset (19,084 videos) includes view count, like count, comment count, and transcription text. The merged master dataset contains 90,032 posts.

3.2 Virality Labels

A binary label was computed per platform using the 80th percentile engagement rate threshold (viral=1, not viral=0). Instagram: $(\text{likes} + \text{comments}) / \text{followers}$, threshold=0.060. YouTube: $(\text{likes} + \text{comments}) / \text{views}$, threshold=0.057. TikTok: $(\text{likes} + \text{comments}) / \text{plays}$, threshold=0.427. Each platform yielded approximately 20% viral posts.

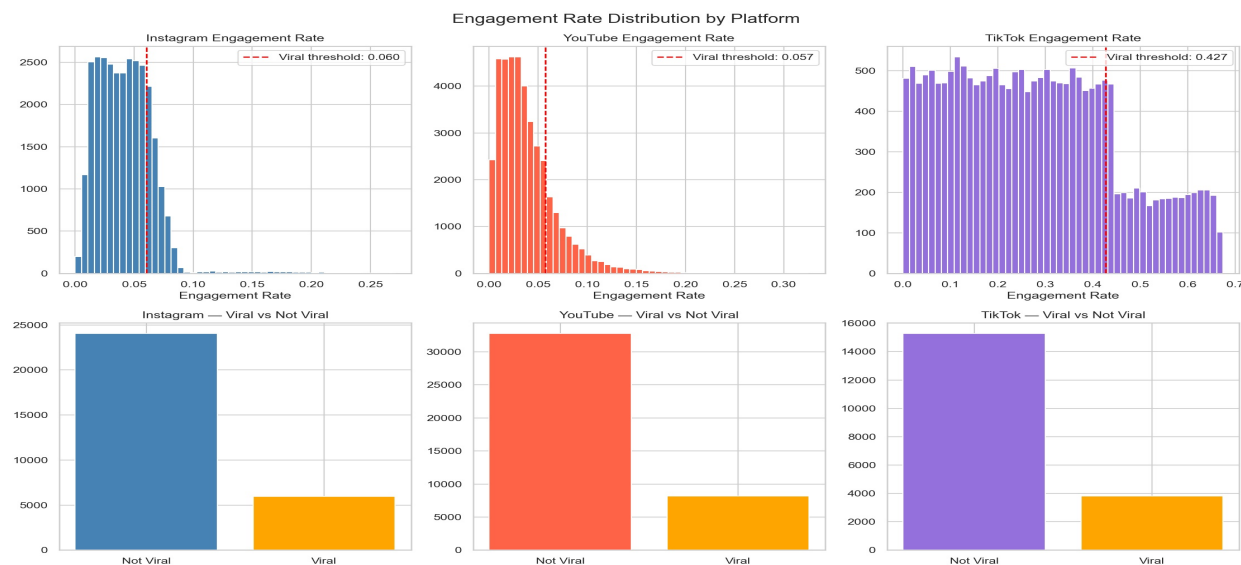


Figure 1: Engagement rate distributions with viral thresholds (top) and viral vs not viral counts (bottom) per platform.

3.3 Features

Feature	Description	Available For
caption_sentiment	VADER compound score (-1 to +1)	YouTube, TikTok
hashtags_count	Number of hashtags	Instagram, YouTube
post_hour	Hour posted (0-23)	Instagram, YouTube
day_of_week	Day of week (0=Mon)	Instagram, YouTube
caption_length	Character count of caption	All platforms
content_category	Category (one-hot encoded)	Instagram, YouTube
platform	Platform indicator (one-hot)	All platforms

Table 1: Features used. TikTok lacked hashtag/timestamp data; neutral defaults were used.

4. Methods

4.1 Pipeline

The pipeline has five stages in Jupyter notebooks: EDA, preprocessing and feature engineering, model training, evaluation, and fairness audit. The master dataset was split 80/20 into training (72,025) and test (18,007) sets using stratified sampling. Features were standardized with StandardScaler.

4.2 Models

Six classifiers were trained: Logistic Regression (balanced class weights, baseline), K-Nearest Neighbors (k=5), Support Vector Machine (RBF kernel, balanced weights), Decision Tree (max_depth=10, balanced weights), Random Forest (100 estimators, balanced weights), and XGBoost (100 estimators, learning_rate=0.1, max_depth=6). All used random_state=42.

4.3 Evaluation Metrics

Models were evaluated on Accuracy, Precision, Recall, F1-score, and ROC-AUC. F1 and ROC-AUC on the viral class were prioritized over accuracy given the 80/20 class imbalance. A fairness audit sliced test predictions by platform and content category.

5. Results

5.1 Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.54	0.23	0.57	0.33	0.61
KNN	0.84	0.68	0.38	0.49	0.76
SVM	0.55	0.26	0.70	0.38	0.66
Decision Tree	0.42	0.25	0.91	0.39	0.68
Random Forest *	0.81	0.53	0.53	0.53	0.80
XGBoost	0.82	0.92	0.11	0.20	0.73

Table 2: All 6 models on test set. * = best model (highlighted).

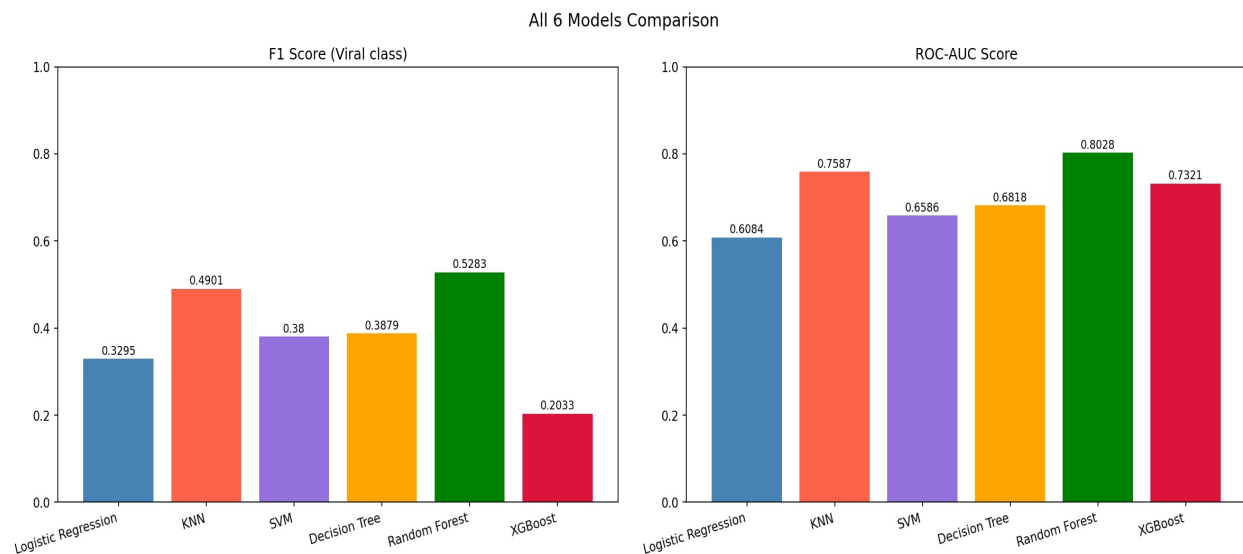


Figure 2: F1 and ROC-AUC comparison across all 6 models. Random Forest achieves the best balance.

5.2 Best Model: Random Forest

Random Forest achieved the best overall performance with $F1=0.53$ and $ROC-AUC=0.80$, balancing Precision and Recall at 0.53 each. XGBoost had the highest precision (0.92) but near-zero recall (0.11), making it impractical. Decision Tree had the highest recall (0.91) but very low precision (0.25). The confusion matrix shows 12,723 true negatives, 1,897 true positives, 1,690 false positives, and 1,697 false negatives.

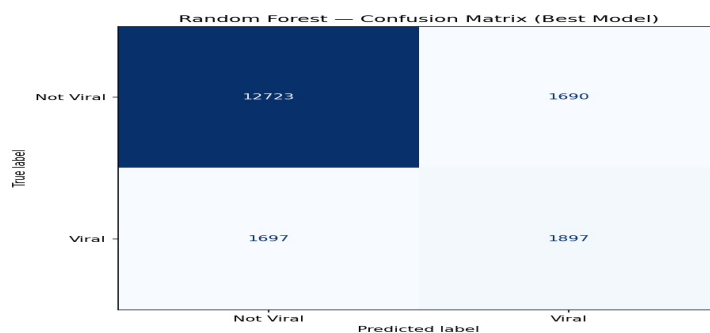


Figure 3: Random Forest confusion matrix on 18,007 test posts.

6. Fairness Audit

6.1 By Platform

Platform	Posts	Viral	Precision	Recall	F1
Instagram	6,133	1,191	0.21	0.05	0.08
YouTube	8,114	1,624	0.80	0.88	0.84
TikTok	3,760	779	0.27	0.53	0.36

Table 3: Platform fairness. Red=worst, Green=best.

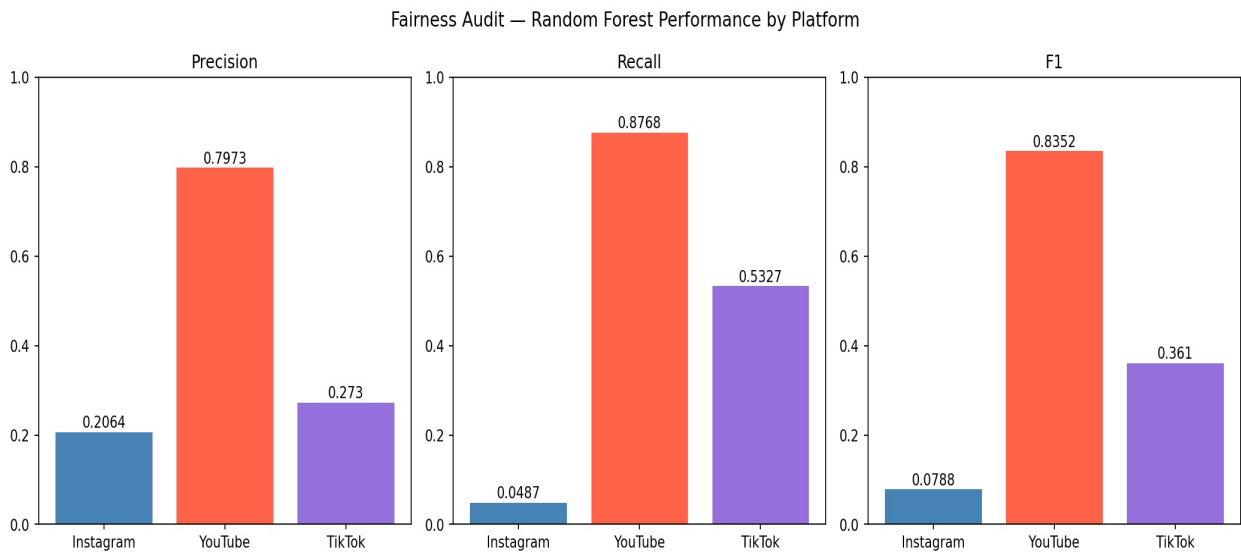


Figure 4: Precision, Recall, and F1 by platform. Massive disparity between YouTube (F1=0.84) and Instagram (F1=0.08).

The results reveal a striking 10x F1 disparity between YouTube (0.84) and Instagram (0.08). YouTube dominates training data (40,949 posts vs 29,999 Instagram) and has richer text features. The model should not be deployed as a fair cross-platform predictor without rebalancing.

6.2 By Content Category

Category	Posts	Viral	Precision	Recall	F1
Fitness	598	111	0.21	0.07	0.11
Music	1,861	524	0.63	0.66	0.64
Comedy	1,293	366	0.87	0.61	0.72
Sports	462	8	0.78	0.88	0.82
Education	315	59	0.91	0.81	0.86
Entertainment	1,984	265	0.84	0.93	0.88

Table 4: Content category fairness. Red=worst.

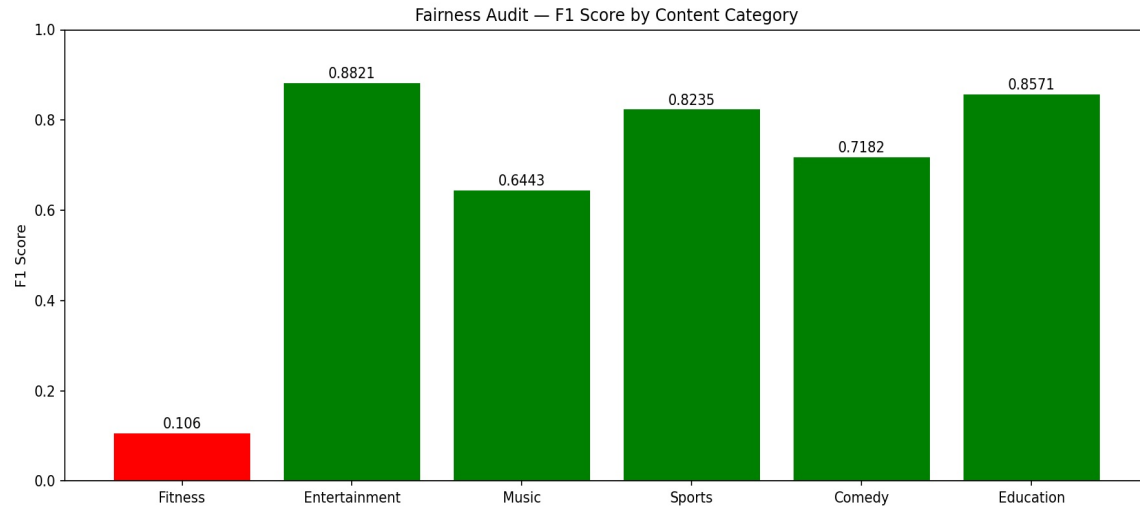


Figure 5: F1 by content category. Fitness (red) severely under-predicted vs Entertainment and Education.

Fitness content shows $F1=0.11$, consistent with Instagram performing poorly since Fitness is predominantly an Instagram category. Entertainment ($F1=0.88$) and Education ($F1=0.86$) perform well due to heavy YouTube representation. Model fairness is closely tied to platform representation in training data.

7. Feature Importance

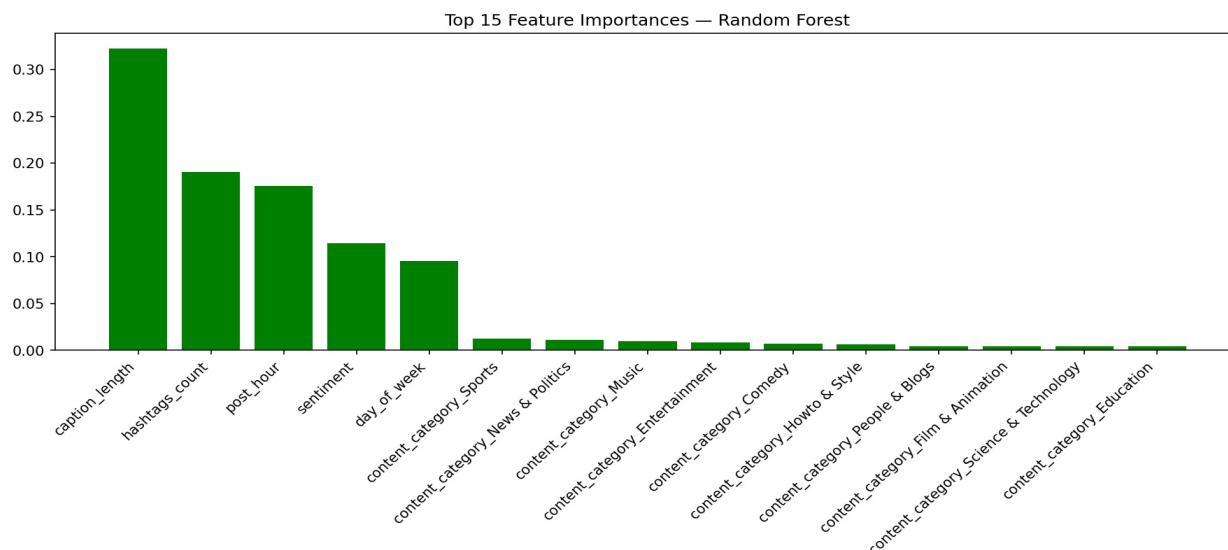


Figure 6: Random Forest feature importances. Caption length dominates (0.32), followed by hashtag count, post hour, sentiment, and day of week.

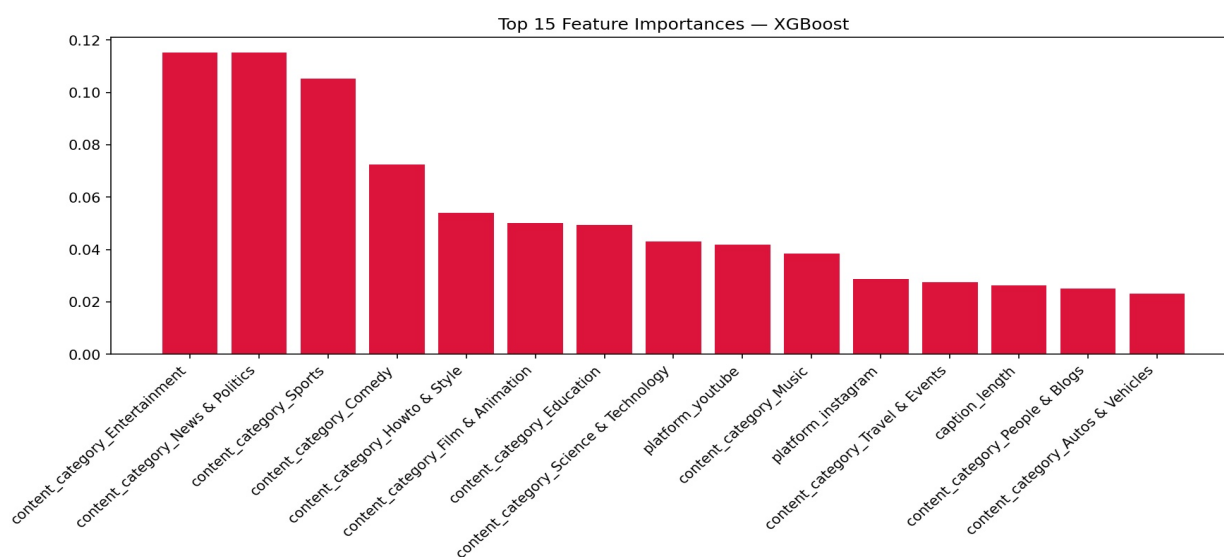


Figure 7: XGBoost feature importances. Content categories dominate: Entertainment, News and Politics, and Sports.

The two models tell complementary stories. Random Forest relies on post metadata: caption length (0.32), hashtag count (0.19), and post hour (0.18). XGBoost relies on content category: Entertainment (0.116), News and Politics (0.115), and Sports (0.105). Both findings are actionable: write longer hashtagged captions, post at optimal times, and choose high-engagement categories.

8. Limitations

First, hyperparameter tuning via GridSearchCV was not performed due to computational constraints on 90,032 rows. Models used carefully chosen defaults; future work could use RandomizedSearchCV or cloud computing.

Second, TF-IDF features were not included. VADER sentiment was used as a lighter text feature. TF-IDF could capture more nuanced vocabulary patterns.

Third, the TikTok dataset lacked hashtag and timestamp columns due to API restrictions, requiring neutral defaults. A more complete TikTok dataset would improve performance.

Fourth, YouTube dominates training data (45% of posts), biasing the model toward YouTube content and causing significant fairness disparities.

9. Conclusion

This project built and evaluated six ML classifiers to predict social media virality using only pre-publication features across Instagram, YouTube, and TikTok. Random Forest achieved the best performance ($F1=0.53$, $ROC-AUC=0.80$) on 18,007 test posts. The fairness audit revealed a 10x F1 disparity between YouTube (0.84) and Instagram (0.08), and Fitness content $F1=0.11$ vs Entertainment $F1=0.88$, driven by training data imbalance. Feature importance showed caption length, hashtag count, and posting time drive Random Forest predictions, while content category drives XGBoost. Future work should use platform-specific models, incorporate TF-IDF, and obtain more complete TikTok data.

References

- Adhikari et al. (2025). Stacking ensemble for social media virality detection. IEEE INCET.
- Silva and Rajapaksha (2023). Instagram virality prediction using caption sentiment. ICATC.
- Vallet et al. (2015). Cross-system virality prediction across YouTube and Twitter. ACM CIKM.
- Roring, Bong and Kulathuramaiyer (2026). Real-time virality model for TikTok and Instagram. IRJMS.
- Noor and Rauf (2025). Comparing ML models for news virality prediction. Social Trends and Policy Review.